Physics Lectures

Quantum Field Theory

Lecture Notes

Quantization, Fock space, propagators, scattering, and renormalization.

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Contents

Preface: What These Notes Are For

Remark

These notes are designed as a continuation of the Classical Field Theory I–II sequence. They do not try to re-teach variational calculus, Lorentz covariance, Noether currents, gauge redundancy, or the classical equations of fundamental fields from the beginning. Instead, they start where the classical notes stop: with fields as dynamical systems ready to be quantized.

Quantum Field Theory After Classical Field Theory

Classical field theory teaches us to describe nature by local fields, actions, symmetries, conservation laws, and equations of motion. Quantum field theory keeps this architecture, but changes the meaning of the basic objects. A classical field configuration is no longer the final description of a physical state. It becomes an ingredient in a quantum theory whose observable content is encoded in states, operators, correlation functions, transition amplitudes, and probabilities.

The guiding idea of the first part of these notes is simple: a free classical field decomposes into normal modes, and each normal mode behaves like a harmonic oscillator. Quantizing the field therefore means quantizing infinitely many oscillators at once. The excitations of these oscillators are interpreted as particles.

What Will Not Be Repeated

The notes assume familiarity with the following ideas from classical field theory:

- the action principle and Euler–Lagrange equations for fields,
- canonical momenta and Hamiltonian density,
- Lorentz covariance and relativistic notation,
- real and complex scalar fields,
- electromagnetic and Yang–Mills gauge fields at the classical level,
- Noether's theorem and conserved currents,
- Green's functions as solutions of linear differential equations.

Whenever one of these topics is used in a quantum calculation, the notes recall just enough of the classical structure to make the calculation readable. The aim is not repetition, but conversion: classical structures are translated into quantum structures.

What Is New in QFT

The genuinely new concepts are the vacuum, Fock space, field operators, propagators, Wick contractions, Feynman diagrams, scattering amplitudes, loop corrections, regularization, renormalization, and scale dependence. These ideas are not decorative additions to classical field theory. They are the mechanisms that make quantum field theory predictive.

A central conceptual warning will appear repeatedly: a Feynman diagram is not a literal picture of tiny particles following tiny trajectories. It is a compact bookkeeping device for terms in a perturbative expansion of correlation functions or scattering amplitudes.

How to Read These Notes

A useful rhythm for reading a QFT calculation is:

$$\text{classical field} \longrightarrow \text{mode expansion} \longrightarrow \text{operators} \longrightarrow \text{states} \longrightarrow \text{correlators} \longrightarrow \text{amplitudes.}$$

At first this may look like a change of notation. It is not. Each arrow changes what is meant by a physical prediction. The purpose of the notes is to make those changes explicit and technically usable.

Chapter 1

From Classical Normal Modes to Quantum Oscillators

Remark

This chapter follows the structure of the course plan in `ideas_Quantum_Field_Theory.md`. Its purpose is to turn the first entry of the table of contents into a readable lecture: a free classical field is decomposed into normal modes, each mode is recognized as a harmonic oscillator, and the oscillator excitations become the first particle states of quantum field theory.

1.1 Starting Point: A Classical Field Decomposed into Modes

Classical field theory describes a field by a function such as $\phi(t, \mathbf{x})$. For a free field, this function can be decomposed into normal modes. This is the same idea that appears in mechanics when a system of coupled oscillators is rewritten in terms of independent normal coordinates.

Begin with a finite chain of N identical masses connected by springs. If $q_j(t)$ is the displacement of the j-th mass, a typical quadratic Lagrangian is

$$L = \sum_{j=1}^{N} \frac{m}{2} \dot{q}_j^2 - \sum_{j=1}^{N} \frac{\kappa}{2} (q_{j+1} - q_j)^2.$$

The variables q_j are coupled. After introducing normal coordinates $Q_r(t)$, one obtains the diagonal form

$$L = \sum_r \left(\frac{1}{2} \dot{Q}_r^2 - \frac{1}{2} \omega_r^2 Q_r^2 \right).$$

The coupled chain is therefore equivalent to a collection of independent oscillators.

A field is the continuum version of this situation. In a one-dimensional chain with spacing a, one may view

$$q_j(t) \approx \phi(t, x_j), \qquad x_j = ja.$$

In the continuum limit, sums over sites become integrals over space, and the normal-mode decomposition becomes a Fourier decomposition.

For a real scalar field in a box of volume V, with periodic boundary conditions, write schematically

$$\phi(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q_{\mathbf{k}}(t) e^{i\mathbf{k}\cdot\mathbf{x}},$$

with the appropriate reality condition for a real field. The allowed momenta are discrete because the field is placed in a finite box. This is a temporary technical convenience; the continuum limit is obtained later by replacing sums with integrals.

Remark

The field is infinite-dimensional, but the free field is not mysterious at this stage. The normal-mode decomposition rewrites it as a collection of independent oscillating degrees of freedom.

1.2 Each Mode as a Harmonic Oscillator

For the free real scalar field, the classical Lagrangian is

$$L = \int \mathrm{d}^3x \left(\frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2 \right).$$

Its equation of motion is the Klein–Gordon equation,

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

After inserting the normal-mode expansion into the free Lagrangian and choosing standard normalizations, the field decomposes into independent oscillator modes,

$$L = \sum_{\mathbf{k}} \left(\frac{1}{2}\dot{q}_{\mathbf{k}}^2 - \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$$

in units with $c = \hbar = 1$.

The canonical momentum of the mode $q_{\mathbf{k}}$ is

$$p_{\mathbf{k}} = \dot{q}_{\mathbf{k}},$$

and the Hamiltonian becomes

$$H = \sum_{\mathbf{k}} H_{\mathbf{k}}, \qquad H_{\mathbf{k}} = \frac{1}{2}p_{\mathbf{k}}^2 + \frac{1}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2.$$

Thus each mode $\mathbf{k}$ is a harmonic oscillator of frequency $\omega_{\mathbf{k}}$.

Example

The classical field is not quantized all at once. First, the free classical field is diagonalized into modes. Then each mode is quantized using the same oscillator algebra known from elementary quantum mechanics.

1.3 Quantizing a Single Oscillator

For one harmonic oscillator, the classical variables q and p become operators $\hat{q}$ and $\hat{p}$ satisfying

$$[\hat{q}, \hat{p}] = i.$$

The Hamiltonian is

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2.$$

Introduce the annihilation and creation operators

$$\hat{a} = \sqrt{\frac{\omega}{2}}\,\hat{q} + \frac{i}{\sqrt{2\omega}}\,\hat{p}, \qquad \hat{a}^\dagger = \sqrt{\frac{\omega}{2}}\,\hat{q} - \frac{i}{\sqrt{2\omega}}\,\hat{p}.$$

Equivalently,

$$\hat{q} = \frac{1}{\sqrt{2\omega}}(\hat{a} + \hat{a}^\dagger), \qquad \hat{p} = -i\sqrt{\frac{\omega}{2}}(\hat{a} - \hat{a}^\dagger).$$

The canonical commutator implies

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

The Hamiltonian becomes

$$\hat{H} = \omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

This is the algebraic form that will be copied for every field mode.

1.4 Creation and Annihilation Operators

The operator $\hat{a}^\dagger$ raises the excitation number of the oscillator, while $\hat{a}$ lowers it. If $|0\rangle$ is the oscillator ground state, then

$$\hat{a}|0\rangle = 0.$$

Repeated action of $\hat{a}^\dagger$ produces excited states,

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \qquad n = 0, 1, 2, \ldots.$$

In a single oscillator problem, this is a convenient way to organize the energy spectrum. In a field theory, the same algebra acquires a new interpretation: creation operators create field quanta.

1.5 Number Operator and Energy Spectrum

The number operator is

$$\hat{N} = \hat{a}^\dagger \hat{a}.$$

It satisfies

$$\hat{N}|n\rangle = n|n\rangle.$$

The Hamiltonian eigenvalue equation is

$$\hat{H}|n\rangle = \omega \left(n + \frac{1}{2} \right) |n\rangle.$$

The integer n counts oscillator excitations. The term $\omega/2$ is the zero-point energy.

For a field, there will be one number operator for each mode,

$$\hat{N}_{\mathbf{k}} = \hat{a}^\dagger_{\mathbf{k}} \hat{a}_{\mathbf{k}}.$$

The eigenvalue of $\hat{N}_{\mathbf{k}}$ counts how many quanta occupy the mode **k**.

1.6 The Vacuum State of One Oscillator

The oscillator vacuum is defined algebraically by

$$\hat{a}|0\rangle = 0.$$

It is not the state of zero energy; its energy is

$$E_0 = \frac{1}{2}\omega.$$

This small fact becomes conceptually important in field theory. A free field has one oscillator for every allowed momentum mode, so the formal vacuum energy is

$$E_{\text{vac}} = \frac{1}{2}\sum_{\mathbf{k}} \omega_{\mathbf{k}}.$$

In the continuum limit this expression diverges. Later chapters will explain normal ordering, regularization, renormalization, and the physical meaning of vacuum energy. At this stage the important lesson is simpler: the quantum vacuum is a state with no excitations, not a state with no structure.

1.7 From One Oscillator to Infinitely Many Oscillators

For the scalar field, each normal mode receives its own creation and annihilation operators,

$$\hat{a}_{\mathbf{k}}, \qquad \hat{a}^\dagger_{\mathbf{k}}.$$

They satisfy

$$[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{k}'}] = \delta_{\mathbf{k},\mathbf{k}'}, \qquad [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}] = 0, \qquad [\hat{a}^\dagger_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{k}'}] = 0.$$

The free-field vacuum is the state annihilated by every annihilation operator,

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \qquad \text{for every} \quad \mathbf{k}.$$

A one-particle state of momentum $\mathbf{k}$ is

$$|\mathbf{k}\rangle = \hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

A two-particle state may be written as

$$|\mathbf{k}_1, \mathbf{k}_2\rangle = \hat{a}^\dagger_{\mathbf{k}_1}\hat{a}^\dagger_{\mathbf{k}_2}|0\rangle,$$

up to normalization conventions.

Definition

The *Fock space* of a free quantum field is the Hilbert space generated from the vacuum by applying creation operators for all allowed modes. It is the natural state space for a theory in which the number of particles is not fixed.

1.8 Field Quanta as Mode Excitations

In the free theory, the Hamiltonian can be written as

$$\hat{H} = \sum_{\mathbf{k}} \omega_{\mathbf{k}} \left(\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} + \frac{1}{2} \right).$$

The first term counts excitations of each mode. The state $\hat{a}^\dagger_{\mathbf{k}}|0\rangle$ has energy $\omega_{\mathbf{k}}$ above the free vacuum and momentum $\mathbf{k}$. It is interpreted as a one-particle state.

The field operator itself is reconstructed from the same mode operators. In a standard continuum normalization,

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + \hat{a}^\dagger_{\mathbf{k}} e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right).$$

The coefficients of the classical mode expansion have become operators. The field operator can annihilate and create quanta because it contains both $\hat{a}_{\mathbf{k}}$ and $\hat{a}^\dagger_{\mathbf{k}}$.

Remark

A field quantum is not a tiny localized lump of classical field. In the free theory, it is a mode excitation created by a creation operator. Localization, wave packets, interactions, and particle detection require additional steps.

1.9 What Is New Compared with Classical Field Theory?

Classical field theory assigns numbers to field configurations. Quantum field theory assigns operators to fields and vectors to states. The basic object is no longer a definite field profile $\phi(x)$, but an operator $\hat{\phi}(x)$ acting on a Hilbert space.

The free quantum field still satisfies the free equation of motion,

$$(\Box + m^2)\hat{\phi}(x) = 0,$$

but the meaning has changed. This is now an equation for an operator-valued distribution, not for a classical function.

The conceptual dictionary of the chapter is:

$$\begin{array}{ccl} \text{classical normal mode} & \longrightarrow & \text{quantum oscillator,} \\ \text{oscillator excitation} & \longrightarrow & \text{field quantum,} \\ \text{many oscillator excitations} & \longrightarrow & \text{Fock-space state,} \\ \text{classical field} & \longrightarrow & \text{field operator.} \end{array}$$

Summary

A free classical field can be decomposed into independent normal modes. Each mode is a harmonic oscillator. Quantizing all of these oscillators produces creation and annihilation operators, a vacuum state, number operators, and Fock space. In the free theory, particles are mode excitations of quantum fields. This is the starting point for propagators, perturbation theory, and scattering.

Chapter 2

Canonical Quantization of the Real Scalar Field

Remark

This chapter turns the oscillator picture of Chapter 1 into the standard canonical quantization of the free real scalar field. The main new object is not one oscillator coordinate, but a field operator $\hat{\phi}(t, \mathbf{x})$ with one canonical commutation relation at every point of space.

2.1 Equal-Time Commutation Relations

We work with a real scalar field in units $c = \hbar = 1$. With metric convention $(+, -, -, -)$, the free Lagrangian density is

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2 = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2.$$

The canonical momentum conjugate to ϕ is

$$\pi(t, \mathbf{x}) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}(t, \mathbf{x})} = \dot{\phi}(t, \mathbf{x}).$$

Classically, the canonical Poisson bracket is

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\}_{\mathrm{PB}} = \delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

Canonical quantization replaces the fields by operators and the Poisson bracket by a commutator:

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

The remaining equal-time commutators are

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0, \qquad [\hat{\pi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = 0.$$

These are the field-theoretic version of the oscillator relation $[\hat{q}, \hat{p}] = i$. Instead of one coordinate and one momentum, there is one canonical pair at every spatial point.

Definition

The *equal-time canonical commutation relations* are the commutators between field and momentum operators evaluated at the same time. They are the canonical starting point of the operator quantization of a bosonic field.

2.2 Field Operator and Momentum Operator

The field operator $\hat{\phi}(t, \mathbf{x})$ is the quantum object corresponding to the classical field $\phi(t, \mathbf{x})$. Its conjugate momentum operator is

$$\hat{\pi}(t, \mathbf{x}) = \partial_t\hat{\phi}(t, \mathbf{x}).$$

The free equation of motion is the Klein–Gordon equation,

$$(\Box + m^2)\hat{\phi}(x) = 0,$$

where $x = (t, \mathbf{x})$. This equation has the same form as the classical field equation, but its meaning is different: it is an equation for an operator-valued object.

The Hamiltonian density of the free field is

$$\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2.$$

After quantization one writes formally

$$\hat{H} = \int \mathrm{d}^3x \left(\frac{1}{2}\hat{\pi}^2 + \frac{1}{2}(\nabla\hat{\phi})^2 + \frac{1}{2}m^2\hat{\phi}^2\right).$$

The word "formally" is important. Products such as $\hat{\phi}(x)^2$ at the same point require care. This issue will return when we discuss normal ordering and renormalization.

2.3 Mode Expansion of the Field Operator

The free Klein–Gordon equation admits plane-wave solutions. The relativistic energy of a mode with spatial momentum $\mathbf{k}$ is

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

A convenient expansion of the real scalar field operator is

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + \hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}}\right).$$

The corresponding momentum operator is obtained by differentiating with respect to time:

$$\hat{\pi}(t, \mathbf{x}) = -i \int \frac{\mathrm{d}^3k}{(2\pi)^3} \sqrt{\frac{\omega_{\mathbf{k}}}{2}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} - \hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}}\right).$$

The normalization factors are chosen so that the canonical equal-time commutation relations take their standard form.

Because the field is real, the operator $\hat{\phi}$ is Hermitian:

$$\hat{\phi}(t, \mathbf{x})^\dagger = \hat{\phi}(t, \mathbf{x}).$$

This is why the expansion contains both $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^\dagger$. The negative-frequency part is the Hermitian conjugate of the positive-frequency part.

2.4 Creation and Annihilation Operators for Momentum Modes

The operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^\dagger$ satisfy

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = (2\pi)^3\delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \qquad [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{p}}^\dagger] = 0.$$

These relations are the continuum analogue of $[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = \delta_{\mathbf{k},\mathbf{p}}$ in a finite box.

The interpretation is inherited from the harmonic oscillator. The operator $\hat{a}_{\mathbf{k}}^\dagger$ creates one quantum of momentum $\mathbf{k}$, while $\hat{a}_{\mathbf{k}}$ annihilates one such quantum. The word "quantum" here means an excitation of the free scalar field mode labelled by $\mathbf{k}$.

Example

If the field is placed in a finite box, the momentum labels are discrete and the commutator is $[\hat{a}_{\mathbf{k}}, \hat{a}^\dagger_{\mathbf{p}}] = \delta_{\mathbf{k},\mathbf{p}}$. In infinite volume, the Kronecker delta is replaced by $(2\pi)^3\delta^{(3)}(\mathbf{k}-\mathbf{p})$.

2.5 Fock Space

The free-field vacuum is defined by

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \qquad \text{for every } \mathbf{k}.$$

The Fock space is built by acting on this vacuum with creation operators. A one-particle state is

$$|\mathbf{k}\rangle = \hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

A two-particle state is

$$|\mathbf{k}_1, \mathbf{k}_2\rangle = \hat{a}^\dagger_{\mathbf{k}_1}\hat{a}^\dagger_{\mathbf{k}_2}|0\rangle,$$

and similarly for higher particle numbers.

For a real scalar field the quanta are neutral bosons. The creation operators commute with one another, so exchanging two creation operators does not change the state. This is the operator origin of Bose symmetry in the free scalar field.

Definition

The *Fock space* of the free real scalar field is the direct sum of sectors with zero, one, two, and more scalar quanta. It is generated from the vacuum by all finite products of creation operators.

2.6 Vacuum, One-Particle States, and Many-Particle States

The vacuum $|0\rangle$ is not a classical field configuration. It is a vector in Hilbert space. It is characterized by the absence of free quanta, not by the absence of operators or fluctuations.

A one-particle state $|\mathbf{k}\rangle$ is an eigenstate of momentum and energy in the free theory. A general one-particle wave packet is a superposition of momentum eigenstates,

$$|f\rangle = \int \frac{\mathrm{d}^3k}{(2\pi)^3} f(\mathbf{k})\hat{a}^\dagger_{\mathbf{k}}|0\rangle.$$

The function $f(\mathbf{k})$ controls the momentum profile of the state. A localized particle-like state is not a single momentum eigenstate, but a wave packet built from many momenta.

Many-particle states are obtained by applying several creation operators. For example,

$$|f, g\rangle = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \int \frac{\mathrm{d}^3p}{(2\pi)^3} f(\mathbf{k})g(\mathbf{p})\hat{a}^\dagger_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{p}}|0\rangle.$$

Because the field is bosonic, the state is symmetric under exchange of the two created quanta.

2.7 Hamiltonian and Momentum Operators

Substituting the mode expansion into the free Hamiltonian gives

$$\hat{H} = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \omega_{\mathbf{k}} \left(\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}} + \frac{1}{2}(2\pi)^3\delta^{(3)}(0)\right).$$

The term proportional to $\delta^{(3)}(0)$ is the infinite-volume version of the zero-point energy. In a finite box, the same expression appears as

$$E_0 = \frac{1}{2}\sum_{\mathbf{k}} \omega_{\mathbf{k}}.$$

The spatial momentum operator is

$$\hat{\mathbf{P}} = \int \mathrm{d}^3x\, \hat{\pi}\nabla\hat{\phi}.$$

In terms of the ladder operators it becomes, after the usual ordering convention,

$$\hat{\mathbf{P}} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \mathbf{k}\, \hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}}.$$

Thus the state $\hat{a}^\dagger_{\mathbf{k}}|0\rangle$ has momentum $\mathbf{k}$ and energy $\omega_{\mathbf{k}}$, exactly as required by the relativistic relation

$$\omega_{\mathbf{k}}^2 = \mathbf{k}^2 + m^2.$$

2.8 Normal Ordering

The infinite constant in the Hamiltonian motivates a useful convention. The normal-ordered form of an operator is obtained by moving all creation operators to the left of all annihilation operators. For the oscillator product one writes

$$:\hat{a}_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{p}}: = \hat{a}^\dagger_{\mathbf{p}}\hat{a}_{\mathbf{k}}.$$

The normal-ordered Hamiltonian of the free scalar field is

$$:\hat{H}: = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \omega_{\mathbf{k}}\hat{a}^\dagger_{\mathbf{k}}\hat{a}_{\mathbf{k}}.$$

It satisfies

$$\langle 0| :\hat{H}: |0\rangle = 0.$$

Normal ordering is not a universal cure for all infinities. It is a clean way to remove the constant vacuum energy of a free field when only energy differences above the free vacuum are relevant. Interacting theories require a more systematic renormalization procedure.

2.9 Zero-Point Energy and Its Interpretation

Every harmonic oscillator contributes $\omega/2$ to its ground-state energy. A field has one oscillator for every momentum mode, so the formal zero-point energy is

$$E_0 = \frac{1}{2}\sum_{\mathbf{k}} \omega_{\mathbf{k}}$$

in a finite box, or

$$E_0 = \frac{V}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3}\, \omega_{\mathbf{k}}$$

in infinite-volume notation. This expression diverges at large $|\mathbf{k}|$.

In non-gravitational scattering calculations, a constant shift of the energy is usually unobservable, so normal ordering is often sufficient at the free-field level. Conceptually, however, the divergence is important. It shows that a field has infinitely many short-distance degrees of freedom. Later chapters will treat such divergences by regularization and renormalization rather than by simply ignoring them.

2.10 Field Operators as Operator-Valued Distributions

The notation $\hat{\phi}(x)$ suggests an operator at a spacetime point. This is useful notation, but it hides a subtle fact: quantum fields are really operator-valued distributions. They become well-defined operators only after being smeared with suitable test functions.

Given a smooth function $f(\mathbf{x})$, define a smeared field at fixed time by

$$\hat{\phi}(f) = \int \mathrm{d}^3x\, f(\mathbf{x})\hat{\phi}(t, \mathbf{x}).$$

This object is better behaved than the formal point operator $\hat{\phi}(t, \mathbf{x})$. Products of fields at the same point, such as $\hat{\phi}(x)^2$, are generally singular and require definition by a limiting or renormalization procedure.

This is not merely a mathematical technicality. The singular behavior of local products is one of the reasons renormalization is unavoidable in interacting quantum field theory.

2.11 Causality and Spacelike Commutators

Relativistic QFT must respect the causal structure of spacetime. For a scalar field, the basic causality condition is that field operators commute at spacelike separation:

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \qquad \text{if} \qquad (x-y)^2 < 0.$$

This property is called microcausality. It means that measurements associated with spacelike separated regions cannot influence one another through the order in which the corresponding operators are applied.

The equal-time commutator

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0$$

is the simplest special case. The nontrivial statement is that the commutator also vanishes for all spacelike separations, not only for equal times.

For the free scalar field, the commutator can be written as a c-number function,

$$[\hat{\phi}(x), \hat{\phi}(y)] = i\Delta(x-y),$$

where Δ is the Pauli–Jordan commutator function. Lorentz invariance and the mass-shell condition imply that $\Delta(x-y) = 0$ outside the light cone. This is the operator statement that the quantized scalar field is compatible with relativistic causality.

Summary

Canonical quantization of the real scalar field begins with equal-time commutators for $\hat{\phi}$ and $\hat{\pi}$. The free field is expanded in momentum modes with creation and annihilation operators. Acting with creation operators on the vacuum builds Fock space, while the Hamiltonian and momentum operators assign energy $\omega_{\mathbf{k}}$ and momentum $\mathbf{k}$ to each one-particle state. Normal ordering removes the free vacuum constant, but the deeper lesson is that local quantum fields are operator-valued distributions and must satisfy microcausality at spacelike separation.

2.12 Chapter Checklist

After reading this chapter, one should be able to:

- write the equal-time commutation relations of the real scalar field;

- derive the canonical momentum from the free scalar Lagrangian;
- explain the mode expansion of $\hat{\phi}$ and $\hat{\pi}$;
- state the commutators of $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^{\dagger}$;
- construct vacuum, one-particle, and many-particle states;
- identify the Hamiltonian and momentum operators in terms of number operators;
- explain what normal ordering does and what it does not do;
- explain why local fields should be treated as distributions;
- state the microcausality condition for a relativistic scalar field.

Chapter 3

The Complex Scalar Field and Antiparticles

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

3.1 Planned Sections

- Quantizing the complex scalar field.
- Two independent sets of ladder operators.
- Particles and antiparticles.
- Charge operator.
- Positive and negative frequency modes revisited.
- Why antiparticles appear naturally.
- Complex scalar propagator.
- Charged quanta and global symmetry.
- Physical interpretation of the Fock space.

Chapter 4

Quantization of the Dirac Field

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

4.1 Planned Sections

- Why fermionic quantization is different.
- Mode expansion of the Dirac field.
- Anticommutation relations.
- Spinor creation and annihilation operators.
- Electron and positron states.
- Charge, momentum, and Hamiltonian operators.
- Positivity of energy and the role of anticommutators.
- Dirac propagator.
- Spin sums and completeness relations.
- Chirality, helicity, and massless fermions in QFT.
- Majorana quantization: conceptual overview.

Chapter 5

Quantization of the Electromagnetic Field

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

5.1 Planned Sections

- Why gauge fields are harder to quantize.
- Physical and redundant degrees of freedom.
- Coulomb gauge quantization.
- Photon creation and annihilation operators.
- Polarization vectors.
- Photon states and helicity.
- Covariant quantization.
- Gauge fixing and the photon propagator.
- Unphysical polarizations and their removal.
- Gauge-invariant observables.
- What is a photon?

Chapter 6

Propagators as Quantum Correlation Functions

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

6.1 Planned Sections

- From classical Green's functions to quantum two-point functions.
- Wightman functions.
- Time-ordered products.
- Feynman propagator.
- Retarded and advanced propagators in QFT.
- The $i\epsilon$ prescription.
- Momentum-space propagators.
- Pole structure and particle interpretation.
- Propagation vs correlation.
- What a propagator actually means.

Chapter 7

The Interaction Picture

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

7.1 Planned Sections

- Schrödinger, Heisenberg, and interaction pictures.
- Free fields as the starting point.
- Interaction Hamiltonian.
- Time evolution operator.
- Dyson series.
- Time-ordered exponential.
- Perturbation theory as an expansion in couplings.
- Vacuum persistence amplitude.
- Adiabatic switching and the meaning of in/out states.
- Why exact interacting QFT is difficult.

Chapter 8

Wick Theorem and Diagrammatics

Remark

This chapter follows the planned QFT table of contents. It is a placeholder to be developed in small increments.

8.1 Planned Sections

- Normal ordering revisited.
- Contractions.
- Wick theorem for scalar fields.
- Wick theorem for fermions.
- Fermionic sign factors.
- From algebra to diagrams.
- External lines, internal lines, and vertices.
- Connected and disconnected diagrams.
- Vacuum bubbles.
- Symmetry factors.
- Interpretation limits of Feynman diagrams.

Chapter 9

Scattering Theory and the S-Matrix

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

9.1 Planned Sections

- In-states and out-states.
- The S-matrix.
- Transition amplitudes.
- Matrix elements and momentum conservation.
- Cross sections.
- Decay rates.
- Lorentz-invariant phase space.
- The LSZ reduction formula.
- From correlation functions to scattering amplitudes.
- Tree-level $2 \to 2$ scattering.
- Decay of an unstable particle.
- From amplitudes to measurable predictions.

Chapter 10

Path Integral for Quantum Mechanics

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

10.1 Planned Sections

- Transition amplitudes as sums over histories.
- Time slicing.
- Classical limit and stationary phase.
- Gaussian integrals.
- Generating functionals in ordinary quantum mechanics.
- Sources and correlation functions.
- Why the path integral is useful for QFT.
- Relation to canonical quantization.

Chapter 11

Path Integral for Scalar Fields

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

11.1 Planned Sections

- Functional integration.
- Free scalar generating functional.
- Sources and derivatives.
- Correlation functions.
- Gaussian functional integrals.
- Perturbation theory.
- Diagrammatic expansion.
- Connected functional.
- Effective action.
- Classical field as expectation value.
- One-particle-irreducible functions.

Chapter 12

Fermionic Path Integrals

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

12.1 Planned Sections

- Grassmann variables.
- Berezin integration.
- Gaussian integrals for Grassmann variables.
- Fermionic generating functional.
- Dirac propagator from the path integral.
- Fermion loops and minus signs.
- Determinants from integrating out fermions.
- Majorana fermions in the path integral.
- Physical interpretation and common pitfalls.

Chapter 13

Gauge Fields in the Path Integral

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

13.1 Planned Sections

- Overcounting gauge-equivalent configurations.
- Gauge fixing in the functional integral.
- Faddeev–Popov method.
- Ghost fields.
- Why ghosts are not physical particles.
- Gauge-fixed photon path integral.
- Gauge-fixed Yang–Mills path integral.
- First introduction to BRST symmetry.
- Gauge independence of physical observables.
- Meaning of gauge fixing in quantum theory.

Chapter 14

ϕ^4 Theory as the First Interacting QFT

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

14.1 Planned Role

This chapter will introduce the first interacting scalar quantum field theory and the basic structure of perturbative Feynman rules, loop corrections, counterterms, and renormalized parameters.

Chapter 15

Yukawa Theory

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

15.1 Planned Role

This chapter will study a scalar field coupled to fermions, Yukawa vertices, simple scattering and decay processes, and the low-energy interpretation of scalar exchange.

Chapter 16

Quantum Electrodynamics

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

16.1 Planned Role

This chapter will introduce QED as the quantum theory of charged matter and light, including Feynman rules, spinor chains, Ward identities, and standard tree-level processes.

Chapter 17

Loop Effects in QED

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

17.1 Planned Role

This chapter will cover radiative corrections in QED, including self-energies, vacuum polarization, vertex corrections, charge renormalization, infrared divergences, and inclusive observables.

Chapter 18

Regularization

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`. Detailed lecture text will be added later.

18.1 Planned Role

This chapter will explain why divergent loop integrals appear and how cutoff, Pauli–Villars, dimensional regularization, and subtraction schemes organize intermediate infinities.

Chapter 19

Renormalization

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

19.1 Planned Role

This chapter will develop the standard QFT treatment of renormalized masses, fields, couplings, counterterms, and physical predictions.

Chapter 20

Renormalization Group

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

20.1 Planned Role

This chapter will organize the scale dependence of quantum field theories and introduce running couplings, beta functions, fixed points, and effective descriptions.

Chapter 21

Effective Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

21.1 Planned Role

This chapter will introduce effective field theory, separation of scales, matching, power counting, symmetries, operator bases, decoupling, and naturalness.

Chapter 22

Non-Abelian Gauge Theory as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

22.1 Planned Role

This chapter will develop quantum non-Abelian gauge fields, self-interactions, gauge-fixed Feynman rules, ghosts, Ward-type identities, and unitarity checks.

Chapter 23

BRST Symmetry

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

23.1 Planned Role

This chapter will introduce BRST transformations, ghost number, nilpotency, physical states as cohomology, gauge independence, and Slavnov–Taylor identities.

Chapter 24

Quantum Chromodynamics

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

24.1 Planned Role

This chapter will introduce QCD field content, color charge, QCD Feynman rules, gluon self-interactions, asymptotic freedom, confinement, jets, and perturbative versus nonperturbative regimes.

Chapter 25

Symmetries in Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

25.1 Planned Role

This chapter will translate classical symmetries into quantum operators, conserved charges, Ward identities, current correlation functions, selection rules, and constraints on observables.

Chapter 26

Anomalies

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

26.1 Planned Role

This chapter will study symmetries that fail after quantization, chiral currents, triangle diagrams, path-integral measure effects, gauge anomalies, anomaly cancellation, and physical consequences.

Chapter 27

Vacuum Structure and Nonperturbative Effects

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

27.1 Planned Role

This chapter will discuss the quantum vacuum, degenerate vacua, tunneling, instantons, theta vacua, false vacuum decay, semiclassical approximation, and nonperturbative contributions.

Chapter 28

The Quantum Electroweak Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

28.1 Planned Role

This chapter will introduce the quantum electroweak sector, gauge fixing after symmetry breaking, massive vector boson propagators, currents, weak processes, and the low-energy Fermi theory.

Chapter 29

The Standard Model Lagrangian as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

29.1 Planned Role

This chapter will organize the Standard Model field content, representations, gauge interactions, Yukawa sector, Higgs sector, flavor mixing, neutrino masses, anomaly cancellation, and EFT viewpoint.

Chapter 30

Selected Standard Model Processes

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

30.1 Planned Role

This chapter will connect QFT calculations with representative Standard Model predictions and collider observables.

Chapter 31

Euclidean Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

31.1 Planned Role

This chapter will introduce Wick rotation, Euclidean correlation functions, reflection positivity, statistical field theory connections, finite temperature, boundary conditions, and Matsubara frequencies.

Chapter 32

Lattice Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

32.1 Planned Role

This chapter will introduce lattice regularization, scalar fields on a lattice, gauge fields on links, Wilson action, Wilson loops, fermion difficulties, Monte Carlo sampling, and the continuum limit.

Chapter 33

Large-N and Semiclassical Methods

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

33.1 Planned Role

This chapter will introduce large-N counting, planar diagrams, saddle-point methods, semiclassical expansion, collective fields, solitons, instanton calculus, and reliability of approximations.

Chapter 34

Conceptual Synthesis I

34.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 35

Conceptual Synthesis II

35.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 36

Summary: The Architecture of Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

36.1 Planned Role

This chapter will summarize quantization, Fock space, propagators, interactions, renormalization, gauge theory, anomalies, nonperturbative effects, the Standard Model, and open directions.

Appendix A

Quantization Dictionary

Remark

This appendix records the translation rules that are used repeatedly in the main text. They should not be treated as mechanical replacements valid in every context. They are starting points that must be checked against constraints, symmetries, domains of operators, and gauge redundancies.

A.1 Classical Mode to Quantum Oscillator

A classical normal mode with amplitude $q_{\mathbf{k}}(t)$ and frequency $\omega_{\mathbf{k}}$ is quantized like a harmonic oscillator. Schematically,

$$q_{\mathbf{k}}, p_{\mathbf{k}} \quad \longrightarrow \quad \hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}}, \qquad [\hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}'}] = i\delta_{\mathbf{k}\mathbf{k}'}.$$

The oscillator is then rewritten in terms of ladder operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}^\dagger_{\mathbf{k}}$.

A.2 Poisson Bracket to Commutator

The canonical rule is

$$\{A, B\}_{\text{PB}} \quad \longrightarrow \quad \frac{1}{i}[\hat{A}, \hat{B}].$$

For fields, the equal-time canonical relation is

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

A.3 Classical Field to Field Operator

A classical field $\phi(x)$ becomes an operator-valued distribution $\hat{\phi}(x)$. The word "distribution" is important: products of field operators at the same spacetime point are generally singular and require a careful definition.

A.4 Classical Green's Function to Propagator

A classical Green's function solves a differential equation with a delta-function source. A quantum propagator is a vacuum expectation value of a time-ordered product,

$$\Delta_F(x - y) = \langle 0 | T\{\hat{\phi}(x)\hat{\phi}(y)\} | 0 \rangle.$$

It is related to Green's functions, but its physical meaning is quantum correlation, not merely classical signal propagation.

A.5 Classical Symmetry to Quantum Ward Identity

A classical continuous symmetry gives a conserved current. In the quantum theory, the same symmetry constrains correlation functions and scattering amplitudes through Ward identities. If the symmetry fails after quantization, the failure is called an anomaly.

Appendix B

Reference B

Placeholder appendix.

Appendix C

Reference C

Placeholder appendix.

Appendix D

Reference D

Placeholder appendix.

Appendix E

Reference E

Placeholder appendix.

List of Figures